MILITARY SPECIFICATION

YOKE ASSEMBLY: 10540507

This amendment forms a part of Military Specification MIL-Y-60291A(MU) dated 18 June 1970.

Paragraph 2.1: Add the following drawing:

"F11744731 Prism, Porro Assembly"

Paragraph 3.5.4, line 4: Delete "as read on the knob scale of the yoke assembly."

Paragraph 4.5.2.3.2: Delete and substitute:

"4.5.2.3.2 Telescope assembly.- The special test equipment utilized for testing the yoke assembly to telescope assembly shall conform to Drawing F10544048 with Porro Prism F11744731 mounted in the 1 1/8 inch diameter holes to permit utilization of the adjusting screws to adjust the vertical axis of the porro reflection and the horizontal axis of the front face reflection with respect to the autocollimator."

Paragraph 4.5.2.3.3: Add the following paragraph:

"4.5.2.3.3 Autocollimator.- An autocollimator with appropriate projection and eyepiece reticle patterns for measuring the mechanical accuracies of the yoke assembly shall be utilized with the test equipment. The optical axis of the autocollimator shall be positioned parallel to the plane of the mounting pads of the adapter mount and such that the porro prism is in the field of view of the autocollimator when the prism is aligned perpendicular to the optical axis of the auto-collimator. The aperture of the autocollimator shall be large enough so as to obtain a porro reflection throughout an excursion of at least ± 250 mils in elevation. The vertical reticle line of the projection pattern shall be aligned perpendicular to the plane of the mounting pads."

Paragraph 4.6.2.4: Delete and substitute:

"4.6.2.4 Elevation backlash (yoke assembly to telscope assembly)- Perform this test utilizing the autocollimator and with the telescope assembly secured in the yoke assembly. Position the elevation knob of the yoke assembly to its midpoint of travel with backlash included for the direction of travel. Observe through the autocollimator and establish coincidence of the horizontal zero reference eyepiece reticle line with the horizontal front surface image line and also establish coincidence of the vertical zero reference eyepiece reticle line with the vertical porro image line. Rotate

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the elevation knob clockwise at least 500 mils, then with backlash included, return the knob to its initial midpoint without overtravel. Record the vertical deflection of the horizontal front surface image line from its initial zero reference position. This deflection is the amount of backlash and shall not exceed the tolerance specified in 3.5.4, as measured with the autocollimator."

Paragraph 4.6.2.4.1: Delete and substitute:

"4.6.2.4.1 Elevation backlash-operating temperature.- The following test shall be performed with the yoke assembly positioned in a temperature test chamber that conforms to the "Test Facilities" requirements of MIL-F-13926 and conditions of 3.4.2. The test chamber shall be provided with an optically corrected window to permit viewing the porro prism. The test shall be performed utilizing the test procedures specified in 4.6.2.4, except the test chamber shall be designed to permit rotation of the elevation knob and viewing the elevation knob scale to determine conformance with 3.5.4. During this test at low and high temperature, the window of the test chamber shall be frost and moisture free respectively so as not to influence any mechanical measurement required for backlash as specified in 3.5.4, as measured with the autocollimator."

Paragraph 4.6.2.5: Delete and substitute:

"4.6.2.5 Lateral displacement.- Conduct this test from the zero position established in 4.6.2.4. Rotate the elevation knob at least two revolutions clockwise and two revolutions counterclockwise. The lateral displacement between the vertical line of the porro reflection and the reference zero position shall not exceed the tolerance specified in 3.5.5 as measured with the autocollimator."

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